Process Mathematics: Classical Analysis, Physics, and Data Compression Without Completed Infinity

A Formal Framework in Rocq from the Theory of Systems

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Abstract

We develop a formal framework for mathematics, physics, and data compression in which infinity is treated as a property of processes rather than completed objects. The central construction is `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$—a function from natural numbers to rationals—which replaces the real number line $\mathbb{R}$.

Within this framework, formalized and fully machine-checked in the Rocq proof assistant (version 9.0.1), we establish:

1. **Classical analysis** (21 theorems including IVT, EVT, Bolzano–Weierstrass, Heine–Borel, FTC, MVT, Taylor, Picard–Lindelöf) without the Axiom of Infinity or Axiom of Choice.
2. **Proof-theoretic strength**: ordinals to ε_0 with well-foundedness *proved* (not assumed), ATR_0 eliminated as a definition, Π^1_1 comprehension collapsed to arithmetic under P4.
3. **Physics from three formulas**: every physical process on a distinction graph is exactly three E/R/R formulas—equation of motion (Rules/L5), spectral decomposition (Roles/L4), field on graph (Elements/L1)—instantiated for sound, light, quantum mechanics, thermodynamics, gravity, and the Standard Model.
4. **Born rule = Parseval theorem**: quantum measurement probabilities are identical to spectral energy fractions. Not analogy—identity through the same formula $P(k) = |A_k|^2$.
5. **Verified data compression**: a GFT-based pipeline achieving 300–600× compression on IoT sensor data, where the compression algorithm *is* the physics simulator.
6. **γ-unification**: decoherence, damping, and compression loss are the *same* equation $A(t{+}1) = (1{-}\gamma)\,A(t)$ with one parameter $\gamma \in [0, 1]$ spanning quantum to classical.

The complete formalization comprises **21,500+ machine-verified theorems** (QED) across **1,480+ files** with **0 Admitted** and only **2 axioms** (classical logic and constructive witness extraction).

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1 Introduction

1.1 The Problem

Standard mathematical analysis is built on the real number line $\mathbb{R}$, conceived as a completed, uncountable set. This construction requires the Axiom of Infinity (to assert the existence of $\mathbb{N}$ as a completed set) and typically the Axiom of Choice (for many existence theorems). The Theory of Systems [1] offers an alternative: derive all of mathematics from the single principle $A =$ exists (something exists), with infinity treated as a property of *processes* (Principle P4), not of completed objects.

An earlier version of this paper [2] demonstrated five classical theorems (IVT, EVT, BW, Heine–Borel, and a mass gap criterion) formalized in Rocq with 7,767 theorems across 353 files. Since then, the project has grown to **21,500+ theorems across 1,480+ files**—expanding from analysis into physics, information theory, and data compression.

1.2 What Is New

This paper extends the original framework in five directions:

1. **Proof-theoretic analysis** (§3): We reach ordinal ε_0 explicitly, prove well-foundedness as a theorem, and show that ATR_0 and Π^1_1-comprehension are eliminable under P4.
2. **Process completeness** (§5): A formal metric on `RealProcess` with diagonal-construction completeness, closing the gap between "Cauchy processes exist" and "Cauchy processes form a complete space."
3. **Physics as E/R/R** (§8): Every physical process is exactly three formulas—field (Elements), spectrum (Roles), evolution (Rules)—with seven verified instantiations from acoustics to the Standard Model.
4. **Born = Parseval** (§9): The Born rule of quantum mechanics is identified with Parseval's theorem, and the quantum–classical transition is parameterized by a single coupling constant γ.
5. **Verified compression** (§10): A practical data compression pipeline derived from the physics framework, achieving 300–600× compression on IoT sensor data.

1.3 Structure of the Paper

§2 defines the universal type and P4 as prohibition. §3 establishes proof-theoretic strength. §4 presents classical theorems as process constructions. §5 proves process completeness. §6 maps our results to Wiedijk's 100 theorems. §7 develops Fourier analysis over $\mathbb{Q}$. §8 presents three E/R/R formulas for physics. §9 establishes Born = Parseval and γ-unification. §10 demonstrates verified compression. §11 updates the Process Mass Gap criterion. §12 discusses scope and limitations. §13 gives formalization statistics.

2 The Universal Type and P4 as Prohibition

2.1 RealProcess

Definition 2.1 (RealProcess)**.** A *real process* is a function $R : \mathbb{N} \to \mathbb{Q}$. We write `RealProcess` $:= \mathbb{N} \to \mathbb{Q}$.

Definition 2.2 (Cauchy condition)**.** A real process R is *Cauchy* if for every $\varepsilon \in \mathbb{Q}$, $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$:

$$|R(m) - R(n)| < \varepsilon.$$

In Rocq:

```
Definition RealProcess := nat -> Q.

Definition is_Cauchy (R : RealProcess) : Prop :=
  forall eps : Q, 0 < eps ->
    exists N : nat, forall m n : nat,
      (N <= m)%nat -> (N <= n)%nat ->
      Qabs (R m - R n) < eps.
```

Remark 2.3. Under P4, the Cauchy process *is* the real number. There is no separate limit object—the process itself is the mathematical entity.

2.2 P4 as Prohibition

The earlier paper presented P4 as a *reinterpretation*: infinity can be replaced by processes. We now establish the stronger result: P4 is *inconsistent* with completed infinities.

Definition 2.4 (Completed infinite set)**.** $\mathrm{CompletedInfSet}(S) :\Leftrightarrow \forall n : \mathbb{N},\ S(n)$.

Definition 2.5 (P4 stage-bounded actuality)**.** $\mathrm{P4_bounded}(\mathrm{actual}) :\Leftrightarrow \forall \mathrm{stage},\ \exists \mathrm{bound},\ \forall n,\ \mathrm{actual}(\mathrm{stage}, n) \Rightarrow n \leq \mathrm{bound}$.

Theorem 2.6 (P4 prohibits completed infinity)**.** $\mathrm{CompletedInfSet}(S) \wedge \mathrm{P4_bounded}(\mathrm{actual}) \wedge \mathrm{bridge}(S, \mathrm{actual}) \Rightarrow \bot$.

Proof. P4 gives a bound B for stage 0. CompletedInfSet gives $S(B{+}1)$. The bridge makes $B{+}1$ actual at stage 0. But $B{+}1 > B$, contradicting the bound. □

Rocq reference: `P4CompletedInfinity.v`, theorem `completed_inf_contradicts_P4`, QED.

Prohibition vs. Reinterpretation

Reinterpretation: P4 provides an *alternative* to the Axiom of Infinity.
Prohibition: P4 is *inconsistent* with completed infinity.
Prohibition implies reinterpretation, but not vice versa. We prove the stronger claim.

Three consequences (`P4ProhibitsAC.v`, `P4ProhibitsImpredicative.v`):

1. Full AC on $\mathbb{N}$ produces a completed infinite object (the choice function) $\Rightarrow$ inconsistent with P4.
2. Impredicative definitions require quantifying over a completed totality $\Rightarrow$ blocked by P4 + P1.
3. Potential infinity ("always more, never all at once") is *compatible* with P4.

3 Proof-Theoretic Strength

3.1 Ordinals to ε_0

We define ordinals as an inductive type in Rocq:

```
Inductive Ord : Set :=
  | OZero : Ord
  | OSucc : Ord -> Ord
  | OLim  : (nat -> Ord) -> Ord.
```

Concrete ordinals:

$$\omega = \texttt{OLim}(\texttt{nat_to_ord}), \tag{1}$$
$$\omega^\omega = \texttt{OLim}(\lambda n.\, \omega^n), \tag{2}$$
$$\varepsilon_0 = \texttt{OLim}(\texttt{omega_tower}), \quad \text{where} \quad \texttt{omega_tower}(0) = \omega,\ \texttt{omega_tower}(n{+}1) = \omega^{\texttt{omega_tower}(n)}. \tag{3}$$

Full arithmetic (addition, multiplication, exponentiation) is defined by structural recursion and verified.

3.2 Well-Foundedness as Theorem

Theorem 3.1 (Well-foundedness of ordinal ordering)**.** *The relation* `ord_lt` *on* `Ord` *is well-founded:* `wf_ord_lt : well_founded ord_lt`*.*

This is a *theorem* (proved by structural induction on `Ord` with an accessibility argument), not an axiom. The proof:

1. **Helper**: `acc_succ`—if x is accessible, then `OSucc`(x) is accessible (by induction on the `Acc` derivation).
2. **Main**: structural induction on `Ord`:
   - `OZero`: nothing is below zero (by inversion).
   - `OSucc`(a): use `acc_succ` on the inductive hypothesis.
   - `OLim`(f): for each k, $f(k)$ is accessible by IH; use `Acc_inv`.

Rocq reference: `TransfiniteInduction.v`, Qed. No additional axioms.

Corollary 3.2 (Transfinite induction)**.** *For any property P:*

$$\big(\forall \alpha.\ (\forall \beta < \alpha.\ P(\beta)) \Rightarrow P(\alpha)\big) \ \Rightarrow\ \forall \alpha.\ P(\alpha).$$

3.3 ATR_0 Eliminated

In reverse mathematics, ATR_0 (Arithmetical Transfinite Recursion) is an axiom asserting that transfinite recursion along any well-ordering produces a definable sequence. Under P4, this becomes a *definition*:

```
Fixpoint iterate_pred (P : (nat -> Prop) -> (nat -> Prop))
    (alpha : Ord) (base : nat -> Prop) : nat -> Prop :=
  match alpha with
  | OZero => base
  | OSucc a => P (iterate_pred P a base)
  | OLim f => fun n => forall k, iterate_pred P (f k) base n
  end.
```

Rocq's termination checker accepts this `Fixpoint` because recursion is structural on `Ord`. The "axiom" becomes a function definition.

Rocq reference: `P4_Eliminates_ATR.v`, 15 Qed.

3.4 Π^1_1 Collapsed

Under P4, every function $\mathbb{N} \to \mathbb{N}$ is a computable program (a natural number encoding). Therefore:

$$\forall f : \mathbb{N} \to \mathbb{N}.\ \varphi(f, n) \quad \equiv \quad \forall c : \mathbb{N}.\ \varphi(\texttt{eval}(c), n).$$

Second-order quantification collapses to first-order (arithmetic).

Rocq reference: `P4_Eliminates_Pi11.v`, 15 Qed.

System	L3 (EM)	Ax. Inf.	AC	Ordinal	Covers
PRA	varies	no	no	ω^ω	primitive recursive
PA	yes	no	no	ε_0	first-order arithmetic
ACA_0	yes	yes	no	ε_0	predicative analysis
ATR_0	yes	yes	no	Γ_0	transfinite recursion
Π^1_1-CA_0	yes	yes	no	$\psi(\Omega_\omega)$	full 2nd-order
ZFC	yes	yes	yes	inaccessible	all standard math
ToS (P4)	**yes**	**no**	**no**	$\boldsymbol{\varepsilon_0}$	**analysis + physics**

Table 1: Position of ToS in the proof-theoretic hierarchy. ToS reaches ε_0 without the Axiom of Infinity, matching PA/ACA_0 in ordinal strength while covering classical analysis (IVT, EVT, BW, HB, FTC, MVT, Taylor, Picard).

3.5 Position in the Hierarchy

4 Classical Theorems as Process Constructions

Every classical existence theorem in analysis has the same structure under P4: construct an explicit process over $\mathbb{Q}$, show it is Cauchy with a computable convergence rate, and verify it satisfies the required property at every stage. We present eight theorems, each decomposed into the E/R/R framework.

4.1 Intermediate Value Theorem

Theorem 4.1 (IVT, Process Version)**.** *Let $f : \mathbb{Q} \to \mathbb{Q}$ be uniformly continuous on $[a, b]$ with $f(a) < 0 < f(b)$. Then there exists a Cauchy process $c : \mathbb{N} \to \mathbb{Q}$ with $c(n) \in [a, b]$ for all n and $f(c(n)) \to 0$.*

Process construction. Define $c(0) = (a + b)/2$. At each step, the interval $[a_n, b_n]$ (with $f(a_n) < 0 < f(b_n)$) is bisected: if $f(c(n)) < 0$, set $a_{n+1} = c(n)$; otherwise $b_{n+1} = c(n)$. The interval width halves:

$$|b_n - a_n| = \frac{b - a}{2^n}.$$

E/R/R analysis.

- **Elements (L1):** midpoints $c(n)$ of nested intervals.
- **Roles (L4):** each $c(n)$ is a rational approximation to the root; the process is Cauchy and bounded.
- **Rules (L5):** bisection—evaluate f at the midpoint, choose the half where the sign change persists (using L3).

Convergence rate: $|c(m) - c(n)| \le (b - a)/2^{\min(m,n)}$ (geometric, $r = 1/2$).
Rocq: `IVT_ERR.v`, 23 Qed. Uses `classic` (L3) for the sign test.

4.2 Extreme Value Theorem

Theorem 4.2 (EVT, Process Version)**.** *Let $f : \mathbb{Q} \to \mathbb{Q}$ be uniformly continuous on $[a, b]$. Then there exists a Cauchy process $x^* : \mathbb{N} \to \mathbb{Q}$ with $x^*(n) \in [a, b]$ such that $f(x^*(n))$ converges to $\sup_{x\in[a,b]} f(x)$.*

Process construction. At stage n, evaluate f on a grid of $2^n + 1$ equally spaced points in $[a, b]$. Set $x^*(n) = \arg\max_k f(a + k(b - a)/2^n)$.

Key feature: When multiple grid points achieve the maximum, L5-Resolution selects the *leftmost*—not by convention but as the unique order-respecting choice (`L5Resolution.v`).

E/R/R analysis.

- **Elements:** grid maximizers $x^*(n)$ on a 2^n-point grid.
- **Roles:** each $x^*(n)$ is the best approximation on its grid; the supremum process $\{f(x^*(n))\}$ is monotonically non-decreasing.
- **Rules:** grid refinement (doubling) + leftmost selection (L5).

Convergence rate: $\sup f - f(x^*(n)) \leq \delta_f(2^{-n})$, where δ_f is the modulus of uniform continuity.
Rocq: `EVT_ERR.v`, 36 Qed.

4.3 Bolzano–Weierstrass Theorem

Theorem 4.3 (BW, Process Version). *Every bounded sequence $s : \mathbb{N} \to \mathbb{Q}$ in $[a, b]$ has a subsequential limit: there exists a nested interval process $[a_n, b_n]$ with $b_n - a_n = (b - a)/2^n$ such that infinitely many terms of s lie in $[a_n, b_n]$.*

Process construction. At each step, bisect the current interval. At least one half contains infinitely many terms of s (by pigeonhole). Choose that half, using L3 (`classic`) to decide which.

E/R/R analysis.
- **Elements:** nested intervals $[a_n, b_n]$.
- **Roles:** each interval contains infinitely many terms of s.
- **Rules:** pigeonhole principle + L3 to choose the half.

Convergence rate: $b_n - a_n = (b - a)/2^n$ (geometric, $r = 1/2$).
Rocq: `BolzanoWeierstrass.v`, 26 Qed.

4.4 Fundamental Theorem of Calculus

Theorem 4.4 (FTC, Process Version). *If $f : \mathbb{Q} \to \mathbb{Q}$ is Lipschitz on $[a, b]$ with antiderivative process $F(n) = \sum_{k=0}^{2^n-1} f(x_k) \cdot \Delta x_n$ (Riemann sums on a 2^n-point grid), then $\{F(n)\}$ is Cauchy and $F'(x) = f(x)$ in the process sense.*

E/R/R analysis.
- **Elements:** Riemann sums $F(n)$.
- **Roles:** each $F(n)$ approximates the integral; the process is Cauchy.
- **Rules:** grid refinement; Lipschitz condition bounds the error:

$$|F(n + 1) - F(n)| \leq L \cdot (b - a)^2/2^{n+1},$$

where L is the Lipschitz constant.

Rocq: `analysis/FTC.v`, 28 Qed.

4.5 Mean Value Theorem

Theorem 4.5 (MVT, Process Version). *If f is differentiable on (a, b) and continuous on $[a, b]$, then there exists a process $c : \mathbb{N} \to \mathbb{Q}$ with $c(n) \in (a, b)$ such that*

$$f'(c(n)) \to \frac{f(b) - f(a)}{b - a}.$$

The process is constructed by applying IVT to $g(x) = f(x) - x \cdot \frac{f(b)-f(a)}{b-a}$, which satisfies $g(a) = g(b)$, then finding the extremum of g via EVT.

Rocq: `MeanValueTheorem.v`, 18 Qed.

4.6 Taylor's Theorem

Theorem 4.6 (Taylor, Process Version). *If f has n derivatives on $[a, b]$, then the Taylor partial sum process*

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

satisfies $|f(x) - T_n(x)| \leq M|x-a|^{n+1}/(n+1)!$, where M bounds $|f^{(n+1)}|$.

Rocq: `TaylorSeries.v`, 18 QED. The bound is verified as a rational inequality.

4.7 Picard–Lindelöf Theorem

Theorem 4.7 (Picard, Process Version). *If $F : \mathbb{Q}^2 \to \mathbb{Q}$ is Lipschitz in the second argument with constant L, and $y_0 \in \mathbb{Q}$, then the Picard iterates*

$$y_{n+1}(t) = y_0 + \int_0^t F(s, y_n(s))\, ds$$

form a Cauchy process with convergence rate $(Lt)^n/n!$.

E/R/R analysis.

- **Elements:** Picard iterates $y_n(t)$.
- **Roles:** each y_n is a better approximation to the ODE solution.
- **Rules:** Lipschitz contraction; the error satisfies $\|y_{n+1} - y_n\| \leq \frac{(Lt)^n}{n!}\|y_1 - y_0\|$.

Rocq: `PicardLindelof.v`, 23 QED.

4.8 The Unified Pattern

Theorem	Elements	Roles	Rules	Rate
IVT	midpoints $c(n)$	Cauchy, bounded	bisection	$1/2^n$
EVT	grid maximizers	monotone, bounded	grid refinement	$\delta(2^{-n})$
BW	nested intervals	shrinking	pigeonhole (L3)	$1/2^n$
Heine–Borel	grid points	covering	Lebesgue number	finite
FTC	Riemann sums	Cauchy	Lipschitz bound	$1/2^n$
MVT	critical pts	extremal	IVT + EVT	$\delta(2^{-n})$
Taylor	partial sums	bounded error	remainder bound	$1/n!$
Picard	iterates y_n	Cauchy	contraction	$(Lt)^n/n!$
Monotone CV	bounded monotone	converging	supremum	$1/n$

Table 2: E/R/R decomposition of nine process theorems. Each follows the same pattern: a process over $\mathbb{Q}$ satisfying a Cauchy/bound condition via a specific interaction rule.

The pattern is uniform: **process + Cauchy + bound = existence**. No completed infinities at any stage. Every convergence rate is an explicit rational expression verified by Rocq's kernel

5 Process Completeness

A standard objection: "Without $\mathbb{R}$, you lose completeness—Cauchy sequences may not converge." We answer: Cauchy processes *do* converge, to a *process*—not to a point in a completed set.

5.1 Metric on RealProcess

We define a stagewise distance between processes:

Definition 5.1 (Stage distance)**.** For processes R, S : `RealProcess` and truncation level N:

$$d_N(R, S) = \sum_{k=0}^{N-1} |R(k) - S(k)|.$$

Properties (all verified in Rocq):

$$d_N(R, R) = 0, \tag{4}$$
$$d_N(R, S) = d_N(S, R), \tag{5}$$
$$d_N(R, S) \leq d_{N+1}(R, S). \tag{6}$$

Under P4, the "full" distance $d(R, S) = \lim_{N\to\infty} d_N(R, S)$ does not exist as a completed object. The sequence $\{d_N(R, S)\}_{N\in\mathbb{N}}$ is itself a process over $\mathbb{Q}$.

Rocq: `ProcessMetricComplete.v`, theorems `proc_dist_N_self`, `proc_dist_N_sym`, `proc_dist_mono`.

5.2 Diagonal Limit Construction

Definition 5.2 (Process-Cauchy)**.** A sequence of processes seq : $\mathbb{N} \to$ `RealProcess` is *process-Cauchy* if for every truncation N and every $\varepsilon > 0$, there exists M such that for all $i, j \geq M$:

$$d_N(\mathrm{seq}(i), \mathrm{seq}(j)) < \varepsilon.$$

Definition 5.3 (Diagonal process)**.** Given seq : $\mathbb{N} \to$ `RealProcess`, the *diagonal process* is:

$$R_\infty(k) = \mathrm{seq}(k)(k).$$

The diagonal takes the k-th value from the k-th process—always a concrete rational.

Theorem 5.4 (Stagewise completeness)**.** *If* seq *is process-Cauchy, then for every stage k and every $\varepsilon > 0$, there exists M such that for all $i \geq M$:*

$$|\mathrm{seq}(i)(k) - R_\infty(k)| < \varepsilon.$$

Proof. Process-Cauchy at truncation $N = k+1$ gives M_k such that $|\mathrm{seq}(i)(k) - \mathrm{seq}(j)(k)| < \varepsilon$ for $i, j \geq M_k$. Take $M = \max(M_k, k)$ and $j = k$: $|\mathrm{seq}(i)(k) - \mathrm{seq}(k)(k)| = |\mathrm{seq}(i)(k) - R_\infty(k)| < \varepsilon$. □

Rocq: `ProcessMetricComplete.v`, theorem `process_completeness`, QED.

Process completeness under P4

The space of Cauchy processes is complete *in the process sense*: every Cauchy sequence of processes converges stagewise to a diagonal process. The limit is itself a process ($\mathbb{N} \to \mathbb{Q}$), not a point in a completed set.
Classical completeness of $\mathbb{R}$ is recovered: take the Cauchy-completeness bridge (`ProcessBridge.v`) to translate any P4 proof into its standard counterpart.

5.3 GenProcess: Observable Streams

The construction generalizes beyond $\mathbb{Q}$:

```
Definition GenProcess (A : Type) := nat -> A.
```

Any stream of observations—rational approximations, measurement outcomes, lattice configurations—is a `GenProcess`. Each prefix is finite:

```
Lemma gp_prefix_length : forall A (p : GenProcess A) n,
  length (gp_prefix p n) = n.
```

This is P4 in action: at every stage, finitely many observations. The "infinite stream" is the process of extending, never the completed totality.

Rocq: `ProcessP4Synthesis.v`, theorem `process_p4_grand_synthesis`

6 Coverage of Wiedijk's 100 Theorems

Freek Wiedijk's "Formalizing 100 Theorems" [7] is a standard benchmark for proof assistants. We have formalized **15 theorems** from the official list plus 7 additional theorems of comparable significance, all without the Axiom of Infinity.

#	Theorem	Rocq File	Special Feature	QED
From Wiedijk's 100 (15 theorems):				
1	$\sqrt{2}$ irrational	`Sqrt2Irrational.v`	infinite descent over $\mathbb{Z}$	14
25	Schroeder–Bernstein	`SchroederBernstein_ERR.v`	depth-based, P4-compliant	16
34	Harmonic series diverges	`HarmonicDiverges.v`	Oresme grouping over $\mathbb{Q}$	21
35	Taylor's theorem	`TaylorSeries.v`	remainder bound	18
39	Fibonacci / golden ratio	`FibonacciGreen.v` + 2 files	Cassini, convergence	34
58	Binomial coefficients	`Combinatorics.v`	over $\mathbb{Q}$	26
63	Cantor's theorem	`ShrinkingIntervals.v`	nested intervals (167 Qed!)	215
64	L'Hôpital's rule	`LHopital.v`	ε-δ limit	25
69	Greatest common divisor	`GCD.v`	Euclidean algorithm	23
74	Cauchy–Schwarz	`CauchySchwarz.v`	$\mathbb{Q}$-vectors	18
75	Bolzano–Weierstrass	`BolzanoWeierstrass.v`	nested intervals + L3	27
78	Mean value theorem	`MeanValueTheorem.v`	via IVT+EVT	18
79	Intermediate value thm	`IVT_ERR.v`	bisection process	24
80	Fund. thm. of calculus	`analysis/FTC.v`	Lipschitz + Riemann sums	29
86	Lebesgue measure	`LebesgueMeasure.v`	process measure	23
Additional theorems of comparable significance (7):				
–	Extreme value theorem	`EVT_ERR.v`	grid refinement + L5	36
–	Heine–Borel	`HeineBorelComplete.v`	Lebesgue number	30
–	Stone–Weierstrass	`StoneWeierstrass.v`	Bernstein polynomials	26
–	Picard–Lindelöf (ODE)	`PicardLindelof.v`	Lipschitz contraction	24
–	Implicit function thm	`ImplicitFunction.v`	process version	21
–	Fixed point (Banach)	`FixedPoint.v`	contraction mapping	20
–	Monotone convergence	`MonotoneConvergence.v`	bounded monotone seq	15
Total: 22 theorems				**508**

Table 3: Theorems from Wiedijk's 100 list (15) and additional theorems of comparable significance (7), all formalized over $\mathbb{Q}$ without the Axiom of Infinity. The Cantor diagonal entry (215 QED) includes both the diagonal argument and the nested interval uncountability proof.

What is unique. Other formalizations (Lean's `mathlib`: 100/100, Isabelle's AFP: 76/100, Mizar's MML: 69/100) prove these theorems within ZFC or HOL—systems that postulate completed infinities. Our 15 theorems from the list demonstrate that these results *do not require* the

Axiom of Infinity: the same mathematical content is captured by Cauchy processes over $\mathbb{Q}$. The 508 combined QED for these 22 theorems represent approximately 2.4% of our total 21,500+ theorems

7 Fourier Analysis Over $\mathbb{Q}$

Classical Fourier analysis requires $L^2(\mathbb{R})$—a completed Hilbert space. Under P4, we replace it with finite DFT on graphs, where everything is exact rational arithmetic.

7.1 DFT on Cycle Graph

The cycle graph C_N has adjacency eigenvalues $\lambda_k = 2\cos(2\pi k/N)$ for $k = 0, \ldots, N-1$. The DFT decomposes a signal $f : \{0, \ldots, N-1\} \to \mathbb{Q}$ into eigenmodes:

$$\hat{f}_k = \frac{\langle f, \varphi_k \rangle}{\|\varphi_k\|^2}, \quad \text{where } \varphi_k \text{ is the } k\text{-th eigenvector of } C_N.$$

For $N = 4$: eigenvalues $\{2, 0, -2, 0\}$, eigenvectors $\varphi_0 = (1, 1, 1, 1)$, $\varphi_1 = (1, 0, -1, 0)$, $\varphi_2 = (1, -1, 1, -1)$, $\varphi_3 = (0, 1, 0, -1)$. All entries rational. All inner products exact.

Rocq: `FourierBasis.v`, 16 QED.

7.2 Parseval Theorem (Exact)

Theorem 7.1 (Parseval over $\mathbb{Q}$). *For any signal f on C_N:*

$$\sum_{j=0}^{N-1} |f(j)|^2 = \sum_{k=0}^{N-1} |\hat{f}_k|^2 \cdot \|\varphi_k\|^2.$$

This is verified by `vm_compute` for $N = 2$ (general signal, symbolic in a, b) and for $N = 4$ (five concrete test signals). No floating-point—exact $\mathbb{Q}$-arithmetic.

Rocq: `FourierConvergence.v` + `FourierGeneralN.v`, theorem `parseval_N2`.

7.3 Fourier as Process

Under P4, the "infinite Fourier transform" does not exist. What exists is the process $\{\mathrm{DFT}_N\}_{N=1,2,\ldots}$. At each stage N:

- N modes, each with a rational coefficient.
- Parseval holds exactly.
- Convolution theorem holds: $\mathrm{DFT}(f * g) = \mathrm{DFT}(f) \cdot \mathrm{DFT}(g)$ pointwise.

The "continuous Fourier transform" is the process of taking $N \to \infty$—never completed.

7.4 Connection to Cayley Transform

The Cayley transform maps Laplacian eigenvalues to transfer matrix eigenvalues:

$$\mathrm{Cayley}(\lambda) = \frac{4 - \lambda^2}{4 + \lambda^2}.$$

This is the bridge from spectral analysis (Fourier) to dynamics (evolution):

$$G_K(0, j) = \sum_k \mathrm{Cayley}(\lambda_k)^K \cdot \frac{\varphi_k(0) \cdot \varphi_k(j)}{\|\varphi_k\|^2}.$$

The Green function is a *finite sum of rational terms*—no integrals, no completed function spaces.

Rocq: `FourierCayleyConnection.v`, 16 QED

8 Three E/R/R Formulas for Physics

Every physical process on a distinction graph is described by exactly three formulas, corresponding to the three components of E/R/R:

```
Record PhysicalProcess := mkPP {
  pp_N        : nat;                                  (* graph size *)
  pp_evolve   : (nat->Q) -> (nat->Q) -> (nat->Q);  (* Rules: HOW *)
  pp_spectrum: (nat->Q) -> nat -> Q;                  (* Roles: WHY *)
  pp_ground   : nat -> Q;                             (* Elements: WHAT *)
}.
```

8.1 The Three Formulas

1. **E-formula** (Elements, L1): $\varphi : \text{Graph} \to \mathbb{Q}$. The field on the graph—what exists at each vertex or edge.

2. **R-formula** (Roles, L4): $\hat{\varphi}_k = \langle \varphi, \psi_k \rangle / \|\psi_k\|^2$. The spectral decomposition—why each mode is significant (its eigenfrequency, energy, or probability).

3. **R-formula** (Rules, L5): $\varphi(v, t{+}1) = F(\varphi(v, t), \varphi(v, t{-}1), \text{neighbors})$. The evolution equation—how the system changes.

8.2 Seven Instantiations

Domain	Rules (L5)	Roles (L4)	Elements (L1)
Sound	wave eq. ($c^2 = 1/4$)	DFT projection	vertex field (silence)
Light	wave eq. ($c^2 = 1$)	DFT projection	edge field (darkness)
QM	Cayley transfer	Born $\lvert A_k \rvert^2$	mode amplitudes
Thermal	coupling (2nd law)	$A_k^2 \omega_k$	equal amplitudes
Casimir	eigenvalue eq.	$\omega_k/2$	vacuum $[0, 2, 4, 2]$
Gravity	geodesic (static)	degree − avg	degree list $[2, 2, 2, 2]$
SM gauge	$N^2 - 1$ generators	gen. count	roles $[2, 3, 1]$

Table 4: Seven physical domains, one record type. Each row is a concrete `PhysicalProcess` instance verified in Rocq.

Rocq: `PhysicalProcess.v`, 14 QED, including `seven_instances_exist`.

8.3 $\sin^2\theta_W = 3/13$: Zero Free Parameters

The Weinberg angle is derived from pure DOF counting:

$$\sin^2\theta_W = \frac{n_{\text{gauge}}}{n_{\text{gauge}} + n_{\text{metric}}} = \frac{3}{3+10} = \frac{3}{13} = 0.23077.$$

Observed: 0.23122 ± 0.00003. Agreement: **0.04%** with **zero free parameters**.

- $n_{\text{gauge}} = 3$: SU(2) generators that mix with U(1).
- $n_{\text{metric}} = 10$: symmetric metric tensor in 3+1 dimensions, $\binom{4}{2} + 4 = 10$.
- The nested distinction $[2, 3, 1]$ at three depths yields SU(3)×SU(2)×U(1) with 12 generators.

Rocq: `DOFCounting.v`, theorem `sin2_is_3_over_13`; `WZMassRatio.v`, theorem `match_within_1pct`

9 The Crown Jewel: Born = Parseval and γ-Unification

9.1 Born Rule IS Parseval Theorem

The Born rule—$P(k) = |A_k|^2$—is not a quantum postulate. It is Parseval's theorem applied to mode amplitudes.

For a normalized state $\psi = [A_0, \ldots, A_{N-1}]$ with $\sum |A_k|^2 = 1$:

$$\text{Born:} \quad P(k) = |A_k|^2, \quad \sum_k P(k) = 1. \tag{7}$$

$$\text{Parseval:} \quad \text{spectral fraction}(k) = \frac{|A_k|^2}{\sum_j |A_j|^2} = |A_k|^2. \tag{8}$$

These are *the same formula*.

Theorem 9.1 (Born = spectral fraction)**.** *For normalized* $\psi = [3/5,\ 4/5,\ 0,\ 0]$*:*

$$P(0) = (3/5)^2 = 9/25 = \textit{spectral_fraction}(0), \tag{9}$$

$$P(1) = (4/5)^2 = 16/25 = \textit{spectral_fraction}(1), \tag{10}$$

$$P(0) + P(1) = 1 \quad \textit{(Parseval).} \tag{11}$$

Rocq: `BornIsParseval.v`, theorem `born_equals_spectral`, Qed.

Furthermore, **compression error = measurement miss**:

$$\underbrace{\sum_{k \geq M} |A_k|^2}_{\text{compression error (keep } M \text{ modes)}} = \underbrace{1 - \sum_{k<M} P(k)}_{\text{measurement miss (mode 0 only)}}.$$

Rocq: `BornIsParseval.v`, theorem `error_equals_miss`, Qed.

9.2 γ-Unification

Three apparently different phenomena—decoherence, damping, and compression loss—are the *same equation* with one parameter:

$$\boxed{A(t{+}1) = (1 - \gamma) \cdot A(t).}$$

γ	Regime	Interpretation
0	Quantum	amplitude preserved forever (coherent)
$0 < \gamma < 1$	Intermediate	exponential decay: $A(n) = (1{-}\gamma)^n$
1	Classical	instant decoherence ($A \to 0$ in one step)

Three names for one equation:

- **Decoherence**: phase information lost to environment modes.
- **Damping**: vibration energy lost to coupled neighbors.
- **Compression loss**: spectral coefficients discarded by truncation.

```
Lemma three_are_one : forall gamma A,
  decoherence_step gamma A == damping_step gamma A /\
  damping_step gamma A == compression_step gamma A.
Proof. intros. split; reflexivity. Qed.
```

The quantum–classical "boundary" is not a boundary. It is a *parameter* $\gamma \in [0, 1]$.

Rocq: `GammaUnification.v`, theorem `quantum_classical_spectrum`, 10 Qed

10 Practical Application: Data Compression

The E/R/R physics framework has a direct practical consequence: the same three formulas (field, spectrum, evolution) that describe physical processes also serve as a verified data compression pipeline. A GFT-based codec achieves 300–600× compression on IoT sensor data, with every error bound machine-verified in Rocq (74 QED across 8 files) and validated by 87 Python tests. The details are the subject of a companion paper [8]

11 Process Mass Gap

11.1 PMG Criterion

Definition 11.1 (Process Mass Gap)**.** A process $g : \mathbb{N} \to \mathbb{Q}$ has process mass gap ε if:

PMG1. $g(M) \geq \varepsilon > 0$ for all $M \geq 0$ (uniform lower bound).
PMG2. $|g(M{+}1) - g(M)| \leq Cr^M$ for some $C > 0$, $0 < r < 1$ (Cauchy rate).
PMG3. $g(M{+}1) \geq g(M)$ for all M (monotonicity).

PMG is strictly stronger than the standard $\lim g(M) > 0$: a process that is zero at one scale (e.g., $g(5) = 0$) fails PMG1 but satisfies the standard criterion. Under P4, the process *is* the physics at every scale.

11.2 SU(2) at $\beta = 1$

The SU(2) lattice gauge spectral gap at coupling $\beta = 1$ satisfies PMG with:

$$\varepsilon = \frac{289}{384} \approx 0.753, \quad r = \frac{1}{4}, \quad C = 1.$$

Rocq: `gauge/ProcessMassGap.v`, 44 QED

12 What Survives, What Changes

12.1 What Survives

All computable mathematics survives intact. The following are proved in Rocq with 0 Admitted:

- All theorems in Table 2 (9 process theorems).
- 20 theorems comparable to Wiedijk's list (Table 3).
- Fourier analysis, convolution, Parseval—exact over $\mathbb{Q}$.
- Physics: Weinberg angle (0.04% agreement), Casimir effect, Born rule.
- Data compression: 300–600× on IoT data.

12.2 What Is Lost

Constructions that require completed infinities:

- **Non-measurable sets** (Vitali): require uncountable AC to select representatives from cosets of $\mathbb{Q}$ in $\mathbb{R}$. Under P4, $\mathbb{R}$ as a completed set does not exist.
- **Banach–Tarski**: requires non-measurable pieces. Dissolved, not lost.
- **Full Kruskal's tree theorem**: requires Π^1_2 strength. We prove it for specific families (chains, forks, bounded depth) but not the full minimal bad sequence argument.

These are *pathologies*, not useful theorems. No physical prediction, no computational algorithm, and no engineering application depends on non-measurable sets.

12.3 What Is Gained

- **No infinities**: every sum terminates, every limit is a process, every bound is rational.
- **Constructive existence**: every existence proof builds an explicit process.
- **Machine verification**: 21,500+ theorems checked by Rocq's kernel.
- **Practical application**: the same formulas compress data and simulate physics.

13 Formalization Statistics

Domain	Files	QED	Key results
Foundation (L1–L5, P1–P4, E/R/R)	117	1,589	Laws, principles, E/R/R, physics
Process framework (Phases 0–10)	297	4,180	Core type, algebra, topology, ODE
Standard library	627	7,510	Ising, Bloch, entropy, trading
Gauge theory (SU(2), SU(3))	114	2,177	Mass gap, beta function, confinement
Riemann zeta & primes	30	683	$\zeta(-1)$, $\zeta(-3)$, zero counting
Classical analysis	34	590	IVT, EVT, BW, FTC, L'Hôpital
Physics (QM, thermo, gravity)	29	556	Born rule, curvature, cosmology
Experimental verification	12	337	Casimir, Bernoulli, vacuum
Lattice QFT (Weinberg angle)	29	281	$\sin^2\theta = 3/13$, one-loop
Projective systems	7	210	Towers, Hilbert, measures
Acoustics & light	18	192	Oscillation, Maxwell, polarization
Linear algebra & set theory	12	228	Eigenvalues, Kruskal, trees
Architecture of Reasoning	6	117	Fallacy taxonomy, AI safety
Extensions (decoherence, etc.)	26	262	γ-unification, entanglement
Fermions & Higgs	7	69	Dirac, Yukawa, gauge loops
Crown jewel	2	21	Born = Parseval
Total	**1,483**	**21,603**	**0 Admitted, 2 axioms**

Table 5: Formalization statistics by domain (exact counts from codebase). All theorems end with QED (no Admitted). Only two axioms: `classic` (L3, excluded middle) and `L4_witness` (P4, constructive witness). Sum verified: $1{,}589 + 4{,}180 + 7{,}510 + \cdots = 21{,}603$.

Comparison with other proof assistants (approximate, as of April 2026):

Project	Theorems	Wiedijk/100	Logic	Axiom of ∞
Lean `mathlib`	200,000+	100	CIC+quotients	yes
Isabelle AFP	150,000+	76	HOL	yes
Mizar MML	60,000+	69	set theory	yes
ToS (this work)	**21,603**	**15**	**CIC + classic**	**no**

We have fewer total theorems and fewer Wiedijk items than the large libraries. However, we are the only project that operates *without the Axiom of Infinity*—demonstrating that classical analysis does not require completed infinities. Our contribution is not size but *foundations*: we cover physics (Weinberg angle, Casimir, Born rule) and data science (compression) that no other proof assistant library addresses from first principles

14 Conclusion

Classical analysis, physics, and data compression work without completed infinities. The framework `RealProcess` $= \mathbb{N} \to \mathbb{Q}$ with P4 (Finite Actuality) provides:

- All classical analysis theorems (21+), including five from Wiedijk's 100 list, with complete machine verification.

- Proof-theoretic strength reaching ε_0, with ATR_0 eliminated and Π^1_1 collapsed.
- A unified physics framework where every process is three E/R/R formulas.
- The identification Born = Parseval and γ-unification of quantum, thermal, and information theory.
- Practical 300–600$\times$ data compression on IoT sensors.

The 21,500+ machine-verified theorems, 0 Admitted, and 2 axioms demonstrate that the framework is not a philosophical proposal but a *working mathematical system*.

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